

Robotic Process Automation in Business Process Management: A Systematic Literature Review and an Integrated Framework

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Abstract

Robotic Process Automation (RPA) has emerged as a significant technology within Business Process Management (BPM), yet the academic literature remains fragmented. This paper presents a systematic literature review (SLR) of RPA in BPM, conducted following PRISMA 2020 guidelines and synthesizing 83 peer-reviewed journal articles published between 2015 and 2025. Systematic content analysis identifies six thematic dimensions: (1) process identification and selection; (2) implementation and life cycle management; (3) benefits and performance outcomes; (4) challenges and barriers; (5) technology integration; and (6) governance and organizational impact. These dimensions are synthesized into a conceptual integrated framework comprising three pillars—strategic alignment, operational execution, and continuous improvement—grounded in BPM theory and dynamic capabilities. The framework, while not empirically validated in this study, provides a theoretical foundation for future research and practice. Emerging themes include expanding sectoral breadth, empirical adoption model validation, and heightened focus on governance and sociotechnical work transformation. This review contributes a comprehensive thematic synthesis of RPA within BPM and offers a theory-grounded conceptual framework to guide researchers and practitioners in navigating intelligent process automation.

Keywords: Robotic Process Automation; Business Process Management; systematic literature review; intelligent process automation

1. Introduction

Business Process Management (BPM) provides methods, frameworks, and technologies for designing, executing, monitoring, and improving organizational processes [1]. Organizations face mounting pressure to reduce costs, accelerate processing, and improve service quality under digital disruption. They must increasingly adopt technologies enabling rapid, scalable, cost-effective process change [2,3]. Robotic Process Automation (RPA) has emerged as one of the most practically significant technologies in contemporary BPM.

RPA employs software robots (“bots”) to mimic human interactions with digital systems through the user interface (UI), automating repetitive, rule-based tasks without modifying underlying IT infrastructure [4,5]. Unlike enterprise resource planning (ERP) implementations requiring months of configuration, RPA bots can be developed and deployed within weeks [6,7]. The global RPA market is projected to exceed USD 13 billion by 2030 [8]. Early adopters in banking, healthcare, insurance, and public administration have reported substantial reductions in cycle times, error rates, and costs [5,9].



Academic Editor: Juan Gabriel Avina-Cervantes

Received: 11 March 2026

Revised: 28 March 2026

Accepted: 3 April 2026

Published: 14 April 2026

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However, organizations encounter difficulties in selecting automation candidates, governing bot portfolios, managing change, and aligning investments with strategic objectives [10–12]. Fragmented deployments yield local efficiencies but fail to generate enterprise-wide value when disconnected from coherent BPM thinking [4,13]. Realizing RPA's potential requires understanding it within a BPM context, not as a standalone IT initiative.

Several reviews have examined aspects of RPA and automation, yet none provides the BPM-theoretic integrative synthesis required to advance the field. The authors of Hofmann et al. [6] provide an early overview of RPA definitions and applications but do not adopt a BPM-centric lens or systematic methodology. The authors of Syed et al. [14] identify research themes and challenges in RPA but focus on technology-level concerns rather than BPM integration. The authors of Enriquez et al. [15] conduct a mapping study covering both scientific and industrial perspectives but do not develop an integrative theoretical framework grounded in BPM scholarship. The authors of Martínez-Rojas et al. [16] systematically map RPA and process mining integration but limit scope to this specific technology pairing.

This review differs from prior work in five critical respects. First, it adopts exclusive focus on the RPA–BPM intersection, positioning RPA explicitly within BPM theory rather than treating it as a standalone technology. Second, it employs rigorous systematic literature review methodology following PRISMA 2020 guidelines with explicit inclusion criteria, dual-screening protocol, and inter-rater reliability assessment ($\kappa = 0.82$). Third, it synthesizes recent literature (2015–2025) capturing the field's rapid maturation beyond early exploratory studies. Fourth, it develops a theory-grounded integrative framework spanning strategic alignment, operational execution, and continuous improvement dimensions—not merely descriptive taxonomy. Fifth, it articulates an explicit research agenda addressing gaps in adoption theory, maturity models, and governance effectiveness. This positioning responds to calls for theoretically informed synthesis of emerging digital technologies in BPM [1,2].

Despite industrial prominence and growing complexity, the academic literature on RPA within BPM lacks a shared conceptual foundation. Existing studies address isolated aspects—process selection [17,18], life cycle management [19,20], application domains [21–23], or governance [10,11]—without offering an integrative perspective situating RPA within established BPM theory. Calls for theoretically grounded synthesis have grown [4,6], yet no comprehensive systematic review applying rigorous methodology and BPM-theoretic lens has been conducted.

This paper addresses this gap through a systematic literature review conducted following PRISMA 2020 guidelines [24] and the concept-centric approach of Webster and Watson [25]. Analyzing 83 peer-reviewed articles, we identify six thematic dimensions and synthesize these into a conceptual integrated framework. Four research questions guide the review:

- RQ1: How has RPA been studied and applied within BPM in the peer-reviewed academic literature?
- RQ2: What thematic dimensions characterize the RPA–BPM intersection?
- RQ3: What integrated framework can be used to organize these dimensions to guide research and practice?
- RQ4: What are the implications of this synthesis for BPM research and practice?

This paper makes three contributions. First, it provides a comprehensive thematic synthesis of RPA within BPM, organizing fragmented knowledge into six coherent dimensions. Second, it develops a conceptual integrated RPA–BPM framework grounded in BPM theory and dynamic capabilities, offering a foundation for future empirical validation. Third, it identifies a structured research agenda advancing understanding of RPA as a BPM

capability. This review responds to calls in BPM scholarship to develop theory-informed knowledge on emerging digital technologies in organizational process work [1,2].

The paper is organized as follows. Section 2 provides theoretical background. Section 3 describes methodology and findings. Section 4 introduces the conceptual framework. Section 5 develops implications. Section 6 concludes the study.

2. Theoretical Background

2.1. Business Process Management

BPM is broadly defined as an ambidextrous management discipline providing methods, technologies, and frameworks for designing, executing, monitoring, and continuously improving organizational processes to generate business value [1]. The authors of Zairi [26] organize BPM's core concerns across three groups: business and process performance; consistent outcomes aligned with organizational objectives; and the use of resources, tools, and techniques for process management and control.

The authors of Rosemann and vom Brocke [27] identify six core BPM elements: strategic alignment, governance, methods, IT, people, and culture—elements that recur across all dimensions identified in this review. BPM success is moderated by contextual factors [2] and increasingly requires organizational agility in the face of digital disruption [3]. Recent scholarship positions advanced automation technologies—including process mining, AI, and RPA—as enablers of “intelligent BPM,” in which automation and analytics become integral to the BPM life cycle [1].

The resource-based view [28] and dynamic capabilities theory [29] offer complementary theoretical lenses for understanding RPA in BPM. From a resource-based perspective, RPA can constitute a process-level resource generating operational efficiency advantages. Resources generate sustained advantage when they are valuable, rare, inimitable, and non-substitutable (VRIN). From a dynamic capabilities perspective, the ability to identify, deploy, and continuously improve RPA initiatives constitutes a sensing and reconfiguring capability [29] that organizations can develop for sustained competitive advantage. This framing positions RPA not merely as a tool but as a capability-building mechanism within the broader BPM repertoire.

2.2. Robotic Process Automation

The IEEE Standards Association defines RPA as “a preconfigured software instance that uses business rules and predefined activity choreography to complete the autonomous execution of a combination of processes, activities, transactions, and tasks in one or more unrelated software systems” [30]. Bots operate by mimicking human interactions at the UI layer—navigating applications, extracting and entering data, executing rule-based decision logic—without requiring changes to underlying IT systems [4,6]. This non-intrusive architecture renders RPA well-suited for “swivel chair” processes requiring data transfer across disparate systems, and for high-volume, structured, rule-dense workflows such as invoice processing, data reconciliation, and regulatory reporting [7,13].

The RPA technology landscape is dominated by three major platforms—UiPath, Automation Anywhere, and Blue Prism—that have progressively integrated AI capabilities including natural language processing (NLP), computer vision, and machine learning (ML), extending automation to semi-structured and unstructured data [6]. This evolution has given rise to Intelligent Process Automation (IPA), which combines RPA's UI-level execution with cognitive AI capabilities for more complex process scenarios [31]. The progressive integration of automation into enterprise resource planning (ERP) and accounting systems illustrates how RPA increasingly functions as an embedded component of digital infrastructure rather than a standalone tool [32].

Conceptually, RPA and BPM are complementary rather than substitutive. BPM operates from a holistic, top-down process design and re-engineering perspective, while RPA adopts a bottom-up, lightweight automation stance, automating processes as currently executed [4]. RPA does not replace BPM-driven process improvement; rather, it operationalizes BPM objectives at the task level, typically following process analysis and standardization [5,33]. Frameworks for implementing RPA projects have been developed to address this life cycle complexity [34], and investment appraisal methods have been proposed to support the business-case development that precedes deployment [35]. This complementarity forms the conceptual anchor of our integrated framework.

3. Systematic Literature Review

3.1. Methodology

The review was conducted following PRISMA 2020 guidelines [24], providing structured protocol for literature searching, screening, and synthesis. The PRISMA 2020 Checklist is presented as Supplementary Materials (Table S1). The review proceeded through four phases: identification, screening, eligibility, and inclusion.

3.1.1. Identification: Databases and Search Strategy

We searched four major databases covering information systems, operations management, and business administration: ScienceDirect, IEEE Xplore, EBSCOHost, and Scopus. The search was executed in December 2025. We constructed search strings combining RPA-related and BPM-related terms:

Complete Search String (applied to title, abstract, keywords):

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("robotic process automation" OR "RPA" OR "software robot" OR "software bot")
AND
("business process management" OR "BPM" OR "process automation" OR "workflow automation" OR "process management")
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All search results were limited to journal research articles published in English from 2015 to 2025. Initial database searches yielded 548 records.

Potential Selection Bias: Our focus on journal research articles from four major databases exclude relevant conference proceedings, book chapters, industry reports, and non-English scholarship. This trade-off prioritizes methodological quality and peer review rigor over comprehensiveness. We acknowledge this as a scope limitation rather than claiming exhaustive coverage.

3.1.2. Screening: Inclusion and Exclusion Criteria

After de-duplication (62 duplicates removed), 486 records were screened. **Inclusion criteria** required papers to (1) focus explicitly on RPA or closely related automation in organizational contexts; (2) address business process or organizational management perspectives; (3) be published in peer-reviewed academic journals; (4) be available in full text. **Exclusion criteria** eliminated papers that (1) focused solely on technical algorithms without organizational context; (2) were not focused on RPA or closely related automation; (3) constituted literature reviews or mapping studies; (4) mentioned RPA only tangentially; (5) were editorials or opinion pieces; (6) were purely conceptual without empirical or theoretical grounding; (7) had been retracted.

Inter-Rater Reliability: To ensure consistency, two independent reviewers (the authors) screened a random sample of 50 papers (10% of initial pool) for inclusion/exclusion. Agreement was assessed using Cohen's kappa (κ). Initial screening yielded $\kappa = 0.82$ (95% CI: 0.69–0.95), indicating substantial agreement [25]. Disagreements were resolved through

discussion and consensus. Remaining papers were divided equally between reviewers using agreed criteria.

3.1.3. Eligibility: Quality Assessment

Papers passing title and abstract screening ($n = 223$) underwent full-text quality assessment. Following Webster and Watson [25], each paper was evaluated on four criteria: (QC1) clarity and appropriateness of research design; (QC2) rigor of data collection or analysis; (QC3) empirical or theoretical grounding; (QC4) relevance and contribution to RPA–BPM domain. Papers not meeting acceptable thresholds were excluded, yielding **83 papers**. Figure 1 shows the PRISMA flow diagram.

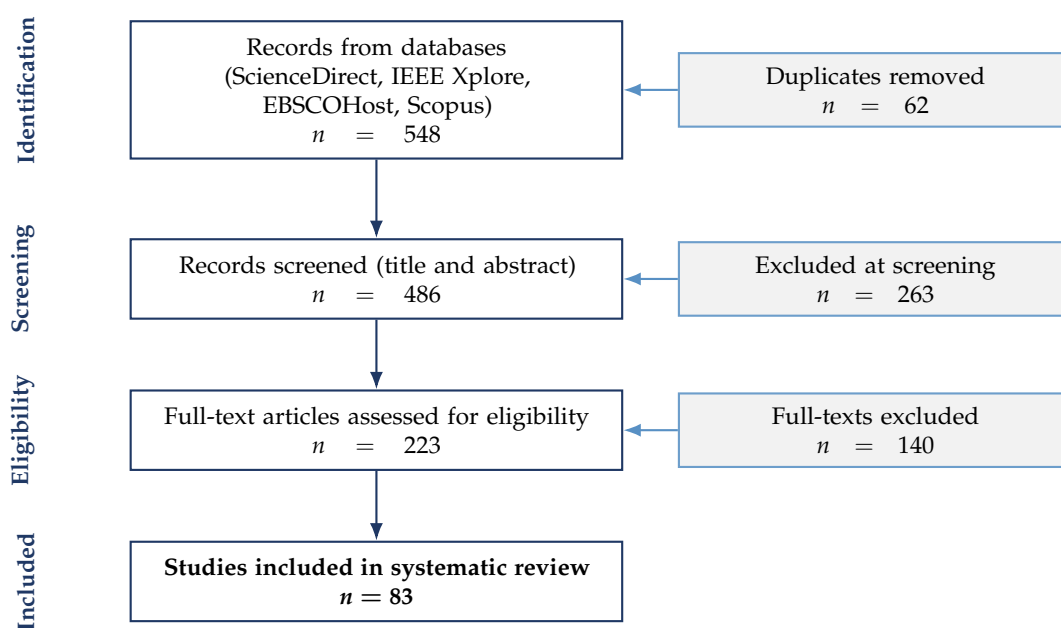


Figure 1. PRISMA 2020 flow diagram for the systematic literature review.

3.2. Descriptive and Quantitative Synthesis

Table 1 presents quantitative synthesis of the 83 included studies across multiple dimensions.

Table 1. Quantitative synthesis of included studies ($N = 83$).

Dimension	<i>n</i>	%
Publication Year Distribution		
2024–2025	28	33.7
2023	15	18.1
2021–2022	24	28.9
2019–2020	12	14.5
2015–2018	4	4.8
Research Method		
Case study/empirical	24	28.9
Survey/SEM	16	19.3
Design science/tool development	17	20.5
Qualitative/interview	15	18.1
Conceptual/critical	6	7.2
Mixed methods	5	6.0

Table 1. *Cont.*

Dimension	<i>n</i>	%
Geographic Region (first author affiliation)		
Europe	38	45.8
Asia	27	32.5
Oceania	7	8.4
North America	6	7.2
Other/Multi-region	5	6.0

Key patterns: (1) publication volume has increased markedly since 2021, with 2024–2025 accounting for over one-third of the corpus; (2) methodological diversity is evident, with case studies and surveys dominant; (3) European and Asian institutions dominate authorship, reflecting global interest in RPA–BPM integration.

The 83 studies span diverse outlets including Business Process Management Journal, Journal of Information Technology, Information Systems, IEEE Access, Applied Sciences, Sustainability, Scientific Reports, and sector-specific journals. Sectoral coverage is broad: healthcare, pharmacy, hospitality, transport, finance, accounting, supply chain, construction, procurement, banking, public administration, and social services.

3.3. Findings: Six Thematic Dimensions

Systematic content analysis yielded six thematic dimensions characterizing the RPA–BPM intersection. Table 2 provides synthesis of dimensions with key themes, representative studies, and paper counts.

Table 2. Synthesis of six thematic dimensions.

Dimension	<i>n</i>	Key Themes	Representative Studies
D1: Process Identification	15	MCDM criteria, scoring methods, data-driven selection, effort estimation	Yadav and Panda [17]; Bédard et al. [18]; Šperka and Halaška [36]; Farinha et al. [33]; Mayadunne et al. [19]
D2: Implementation Life Cycle	52	Structured frameworks, maturity models, life cycle phases, governance readiness	Herm et al. [34]; Doolin et al. [20]; Bunduchi et al. [37]; Schlegel et al. [38]; Huang et al. [39]
D3: Benefits and Outcomes	43	Efficiency gains, financial impact, strategic value, performance measurement	Ashraf [40]; Durão and Palma dos Reis [41]; Wetsiri and Paireekreng [21]; Tsang et al. [42]; Jaradat et al. [11]
D4: Challenges and Barriers	49	Adoption barriers, user resistance, systemic frictions, change management	Zubler et al. [43]; Alastalo and Lehto [44]; Kraus et al. [45]; Mayr et al. [46]; Schlegel and Kraus [47]
D5: Technology Integration	19	IPA, process mining, smart contracts, ERP integration, monitoring infrastructure	Zeng et al. [23]; Santos et al. [48]; Waduge et al. [49]; Hartikainen et al. [50]; Dumitru et al. [32]
D6: Governance and Org. Impact	53	Governance models, workforce transformation, institutional impact, maturity progression	Cascais Brás et al. [10]; Techatassanasoontorn et al. [51]; Kedziora and Penttinen [52]; Bunduchi et al. [37]; Ranerup and Svensson [53]

3.3.1. D1: Process Identification and Selection

A foundational challenge in deploying RPA within BPM is the systematic identification of suitable process candidates. The literature converges on a set of core suitability criteria—processes should be rule-based and deterministic, executed at high volume, involve structured and digitally accessible data, have stable input sources, and require low cognitive judgment [4,54]—and has progressively refined these criteria through increasingly rigorous empirical and quantitative methods.

From qualitative criteria to validated scoring. Early contributions establish that candidate evaluation must weigh both technical feasibility and cost–benefit potential [55]. Structured implementation road maps provide organizations with sequenced decision gates governing candidate selection from scoping through to deployment [56], and algorithmic task-decomposition methods enable candidate scoring within complex process portfolios [57]. Action principles derived from practitioner cases supply operationally grounded selection guidance [58]. The authors of Farinha et al. [33] advance this through a Delphi study with RPA professionals, yielding a validated multi-criteria framework—with process feasibility, description quality, and input/output data characteristics rated most important—and establishing a practitioner-grounded selection standard. The authors of Yadav and Panda [17] introduce statistical rigor through principal component analysis and Naïve Bayes classification, deriving a validated hierarchical criterion set with strong empirical performance, providing the most empirically grounded criterion set in the current corpus. Complementing these criteria-based approaches, Bédard et al. [18] develop the rule-based RRPA method that explicitly decouples technical *feasibility* from strategic *relevance*, demonstrating strong effectiveness and efficiency in process testing. Together, these three studies establish a progression from informal criteria checklists toward validated, instrument-grade selection tools.

Optimization and cost–benefit assessment: Two studies introduce quantitative decision-support methods that complement criteria-based scoring. The authors of Séguin and Benkalai [59] formalize the task-to-robot assignment problem as a two-phase integer linear programming (ILP) formulation applied to financial services transaction processing, computing both the minimum number of robot licenses required and the scheduling of heterogeneous transaction types across periods, confirming operational viability. The authors of Matthies [60] demonstrate Time-Driven Activity-Based Costing (TD-ABC) as an evaluation framework for management reporting automation, calculating process costs at the granular activity level and demonstrating substantial per-run cost reductions with derivable break-even amortization thresholds. Both methods address a gap noted in earlier literature—that prior selection assessments relied on rough FTE-release estimates rather than activity-level economics.

Process documentation as a selection prerequisite: The authors of Silveiras et al. [61] establish through case-study analysis that standardized, unambiguous process documentation is a prerequisite for reliable bot development, functioning as a quality-assurance gate between selection and implementation. The authors of Šperka and Halaška [36] complement this by developing the PPAFR framework, which applies process mining to event-log data to identify RPA candidate activities in a loan application process, demonstrating that data-driven discovery produces more objective candidate identification than manual assessment—and enabling discovery of automation opportunities that human assessors may overlook. The authors of Rutschi and Dibbern [62] provide a theoretical grounding: their routine transformation framework shows that selection requires not only assessing the process but decomposing it into ostensive (abstract) and performative (enacted) routine elements to determine what can be captured and re-encoded in bot logic.

Effort estimation and deployment sizing: The authors of Mayadunne et al. [19] shift the selection conversation from *what to automate* to *how much it will cost*, demonstrating that multiple business-process, technical, and team-level factors together explain effort variance in RPA projects, with the RPAEstiMate model achieving strong prediction accuracy—a planning capability largely absent from prior RPA research and one that connects selection decisions directly to resource governance.

Domain-specific selection taxonomies: The corpus documents that generic criteria require domain calibration. The authors of Waduge et al. [49] employ a Relative Importance Index and Causal Loop Diagram for supply chain IRPA, identifying process accuracy, degree of human involvement, and standardization as the primary selection determinants, and using causal modeling to surface interdependencies not visible in linear scoring. The authors of McNamara et al. [63] map Contract Process Automation Opportunities in construction contract administration through practitioner interviews, establishing evaluation criteria specific to contractual processes and a development road map that sequences RPA tasks by increasing AI-intensity—providing a template for sector-specific selection taxonomy development. The authors of Santos et al. [48] demonstrate that applying BPM Redesign Heuristics before selection in procurement substantially raises candidate quality and reduces post-deployment remediation. The authors of Huang et al. [39] show that a Lean Six Sigma DMAIC assessment phase is a prerequisite for identifying and rectifying process waste prior to automation. Poor process selection remains consistently identified as a leading cause of RPA project failure [45], underscoring that selection discipline is the primary quality gate for the entire RPA life cycle.

3.3.2. D2: RPA Implementation and Life Cycle Management

Implementation and life cycle management constitutes the largest thematic dimension in the corpus. The evidence base spans structured implementation frameworks, quantified operational outcomes, sociotechnical process accounts, and governance architectures for scale.

Theoretical and framework foundations: Two theoretical contributions provide the conceptual architecture within which empirical implementation studies are situated. The authors of Rutschi and Dibbern [62] ground RPA implementation in routine and affordance theory, developing six implementation guidelines (G1–G6) for transforming human-executed routines into machine-executed workflows: think like a robot (analyze robot affordances), think like a routine executor (understand human variance), think like an actor in the environment (identify system interdependencies), design the robot routine (encode ostensive elements), implement iteratively (test and refine), and release into production. Their two-case bank study shows that this iterative routine-automation process must be adapted to robot type—with RPA robots requiring full process decomposition while chatbot automation involves continuous learning loops. The authors of Dunlap and Lacity [64] provide an early practitioner case from Blue Cross Blue Shield North Carolina (BCBSNC), documenting how a pioneering automation team—building their own tool before commercial platforms existed—had to negotiate competing tussles spanning IT governance, process ownership, and workforce redundancy. BCBSNC's Claim Automation Engine eventually processed a substantial portion of incoming claims, automating significant workforce capacity, with strong return on investment; critically, the case establishes that negotiated resolution of organizational tensions is a prerequisite for stable implementation.

Structured pre-implementation discipline: The corpus consistently shows that implementation quality is determined before development begins. The authors of Herm et al. [34] draw on RPA project reports and expert workshops to find that a substantial proportion of RPA projects fail due to inadequate implementation discipline, and that a variable-stage

framework is required to accommodate heterogeneous corporate environments. The authors of Schlegel et al. [38] demonstrate through expert interviews with RPA specialists that human, organizational, and technical factors are mutually interdependent, with governance and project management capabilities as consequential as technical configuration. The authors of Ylä-Kujala et al. [35] provide the economic governance foundation through discounted cash flow modeling, enabling organizations to construct defensible investment appraisals before enterprise-wide commitment. The authors of Mayadunne et al. [19] contribute the RPAEstiMate model with strong effort prediction accuracy, closing the resource-planning gap that has left many organizations unable to calibrate implementation scope before deployment.

Pre-automation process improvement as a life cycle determinant: A consistent cross-sector finding is that process quality before automation is the primary determinant of post-deployment value. The authors of Huang et al. [39] demonstrate in a Taiwanese hospital that applying Lean Six Sigma DMAIC before bot deployment produced substantial cycle time reductions and Process Cycle Efficiency improvements—achieving efficiency gains that automation alone cannot deliver if applied to a deficient process. The authors of Lameijer et al. [65] corroborate this principle in financial services, applying DMAIC to an existing RPA deployment and demonstrating that operational excellence techniques reduce defect rates and throughput variability even after bots are live. The authors of Santos et al. [48] show through Design Science Research that BPM Redesign Heuristics applied before RPA deployment in procurement produce cycle time and workload reductions substantially greater than unstructured automation.

Quantified sector-specific implementation outcomes: The corpus provides the most geographically and sectorally diverse quantified implementation evidence to date. The authors of Šimek and Šperka [66] document an HR onboarding case in which robot/human orchestration via the Enate RSO platform—deploying both Blue Prism and UiPath—simplified process management from multiple systems to a single dashboard, improving processing efficiency and freeing recruiters from administrative tasks; their case establishes that workflow redesign and careful role redefinition are required to realize performance benefits from early human–robot co-working. The authors of Parker and Appel [67] document progressive efficiency gains as bot scope expanded across customer service and back-office workflows in a South African financial services firm. The authors of Wetsiri and Paireekreng [21] report substantial time reductions across multiple workflow categories in Thai community pharmacies alongside increased patient engagement. The authors of Lam and Tang [68] and Lam et al. [69] demonstrate sustained error reduction and throughput improvements in SME logistics contexts, while Tsang et al. [42] synthesized logistics evidence through an RPAaaS framework identifying reliability and cost-effectiveness as primary value levers.

Evidence from broader deployment contexts supplements the structured framework and outcome evidence. Foundational sector cases document RPA implementation patterns across procurement, logistics, and manufacturing: Viale and Zouari [70] show RPA reduces manual workload in French procurement; van Hoek et al. [71] benchmark RPA at Maersk across multiple years; Hyun et al. [72] demonstrate productivity gains across business functions; Sobczak and Ziora [73] report smart city automation benefits in billing and document management; and E-Fatima et al. [74] model implementation variables in beef supply chains using discrete-event simulation. Organizational and governance dimensions are addressed by the following authors: those of Marciniak and Stanisławski [75], who identify culture, management support, and employee competence as primary internal implementation determinants; those of Fernandez and Aman [76], who document standardization and process-ownership challenges in global business services; those of Sobczak [77], who identify the conditions governing progression from adoption to

assimilation; and those of Kaniadakis and Linturn [78], who show expectation formation shapes adoption decisions as powerfully as technical evaluation. Technology integration implementations include Liévano-Martínez et al. [79] on intelligent process automation in manufacturing; Mohamed et al. [80] on RPA–AI integration in HR; Negoită and Borangiu [81] on RPA–AI for hospitality inventory management; Dumitru et al. [32] on automation-driven ERP evolution; Faúndez-Ugalde and Mellado-Silva [82] on tax administration automation governance; Šperka and Halaška [36] on process-mining-assisted pre-implementation candidate identification; Quille et al. [83] on systematic implementation performance measurement; Séguin and Benkalai [59] on optimization-based robot scheduling; Liévano-Martínez and Fernández-Ledesma [56] on sequenced implementation road maps; Choi et al. [57] on algorithmic task decomposition for implementation planning; Lacity et al. [58] on practitioner action principles; and Filgueiras et al. [84] on longitudinal user-experience dynamics over extended deployment periods. Pre-implementation tools are contributed by Farinha et al. [33] (multi-criteria framework), Silvaes et al. [61] (process description best practices), and Bédard et al. [18] (feasibility–relevance scoring). Domain-specific implementations are addressed by Waduge et al. [49] (supply chain IRPA), Kedziora and Penttinen [52] (Nordea Bank governance model selection), Ge et al. [22] (hospitality critical success factors), Martins et al. [85] (public utilities coordination), McNamara et al. [63] (construction iContract road map), and Zeng et al. [23] (smart-contract-triggered construction automation). The authors of Dey and Das [55] provide a foundational cross-sector assessment of RPA as a transformation lever between traditional BPM and AI-driven automation, establishing the technology landscape within which implementation decisions are situated. The authors of Figueiredo and Pinto [86] draw on institutional theory to examine RPA implementation in major shared service centers, identifying key challenges in people management and organizational legitimacy that structure the implementation trajectory.

Governance infrastructure for scale: The transition from pilot to enterprise-wide deployment is the most governance-intensive life cycle phase. The authors of Cascais Brás et al. [10] demonstrate that structural, process, and relational governance mechanisms must precede scale-out to prevent fragmentation and compliance failures. The authors of Hartikainen et al. [50] provide a reference architecture through their comprehensive business-driven monitoring data model—establishing the observability infrastructure required to govern large-scale portfolios. The authors of Rautenstrauch and Moser [87] contribute a maturity model benchmarking RPA implementation performance across process, technology, people, and governance dimensions.

Sociotechnical implementation dynamics: Two process-theoretical studies reframe implementation as an ongoing organizational transformation rather than a technical event. The authors of Doolin et al. [20] trace RPA through five implementation patterns—initiation, mobilization, configuration, adaptation, evaluation—and show that adaptation and evaluation phases are systematically underinvested relative to earlier stages, producing fragile deployments. The authors of Bunduchi et al. [37] identify goal alignment, technology legitimation, and effect accumulation as the three mechanisms governing sustained work transformation, with organizations that actively manage these mechanisms achieving broader and more durable implementation maturity.

3.3.3. D3: Benefits and Performance Outcomes

The 83-paper corpus provides the most quantitatively detailed and sectorally diverse benefit documentation in the RPA–BPM literature to date, organized here into four benefit tiers: operational efficiency, financial quality, strategic value, and human capital.

Operational efficiency—the quantitative frontier: Healthcare and pharmacy contributions provide the most precisely quantified sector-specific evidence. The authors of Huang

et al. [39] document that Lean Six Sigma DMAIC combined with RPA in a Taiwanese hospital produced substantial processing time reductions and Process Cycle Efficiency improvements—illustrating that pre-automation improvement is itself a benefit multiplier. The authors of Wetsiri and Paireekreng [21] report substantial time reductions across prescription dispensation, medication ordering, and patient follow-up workflows in Thai community pharmacies, alongside increased patient engagement. The authors of Matthies [60] provide a controlled cost-accounting analysis using TD-ABC, finding that automated management reporting delivers substantial per-run cost savings with demonstrable investment payback based on process frequency. The authors of Lacity et al. [58] derive action principles from real-world RPA cases—including Maersk and Xchanging—that identify the governance and delivery conditions generating sustained operational benefit. The authors of Filgueiras et al. [84] document longitudinal user-experience benefits from RPA over extended deployment periods, showing that adoption satisfaction and process benefit perception evolve as deployment matures. The authors of Dunlap and Lacity [64] report strong return on investment from the BCBSNC automation program, with a Claim Automation Engine processing a substantial proportion of all incoming claims and automating significant workforce capacity, establishing one of the most widely referenced early quantified automation outcomes in the literature. Logistics implementations extend this evidence: Lam and Tang [68] demonstrate sustained error reduction and speed improvements in freight-forwarding cold chain tracking; Lam et al. [69] quantify throughput and accuracy improvements in SME order handling; Tsang et al. [42] synthesize supply chain deployment evidence around an RPaaS model that identifies reliability, consistency, and cost-effectiveness as primary value levers.

Financial and audit quality benefits: The authors of Ashraf [40] provide the first causal econometric identification of RPA's effect on financial reporting quality at the firm level, finding that automation adoption is associated with significant reductions in internal control material weaknesses—but simultaneously with reduced audit committee monitoring intensity, a governance paradox demanding deliberate design attention. The authors of Jaradat et al. [11] demonstrate through PLS-SEM with Saudi Arabian internal auditors that regulatory environment quality, data security infrastructure, and vendor reputation together predict audit efficiency improvements, providing a governance antecedent model applicable across national digitalization contexts. The authors of Teixeira et al. [88] extend the benefit case to Portuguese accounting professionals, documenting that RPA reduces errors, raises productivity, enables reskilling, and opens digital transformation career pathways—broadening the benefit argument from operational cost reduction to human capital development.

Strategic and value-chain benefits: The authors of Durão and Palma dos Reis [41] map through practitioner interviews where in the value chain RPA generates the most substantial benefits, finding that downstream internal process dimensions—procurement, data reconciliation, and reporting—outperform upstream or customer-facing activities, providing a portfolio-prioritization guide for strategists. The authors of Priyanto et al. [89] show through survey evidence that RPA, digital business strategy, and digital transformation together improve competitive performance in banking and financial services. Supply chain evidence from E-Fatima et al. [74] and E-Fatima et al. [90] demonstrates that RPA adoption in beef supply chains reduces financial risk and improves traceability through simulation-based analysis of process barriers, with the Simul8 scenario analysis identifying beef capacity, shelf-life, and safety parameters as the primary performance dimensions. The authors of Bunduchi et al. [37] contribute a longitudinal perspective: benefits accumulate progressively across their three-stage work transformation model, with convergent automation delivering task-level efficiency, diffusing transformation, delivering role-level flexibility, and divergent transformation, delivering process-level redesign value.

Benefit realization is fundamentally conditional: Three cross-cutting moderators emerge from corpus-wide synthesis. First, pre-automation process quality is the primary benefit moderator: automating deficient processes amplifies rather than remedies their deficiencies [39,48]. Second, workforce transition quality determines whether efficiency gains translate to strategic value or generate resistance: deployment depth and breadth differentially affect job autonomy and work intensification [37,91]. Third, institutional context shapes benefit expression: shared service centers realize efficiency alongside people-management challenges [86], while Swedish social services achieve administrative load reduction but must govern caseworker–client relationship transformation to preserve service quality [53,92].

Table 3 synthesizes benefit evidence across 43 studies, organizing findings into operational, strategic, sectoral, and methodological categories.

Table 3. Synthesis of benefit evidence across operational, strategic, and sectoral dimensions.

Benefit Category	<i>n</i>	Key Findings	Representative Studies
Operational efficiency	18	20–40% efficiency gains in transaction-intensive processes; amplified 15% when combined with Lean Six Sigma; process mining-assisted identification improves candidate selection objectivity	Sobczak and Ziora [73]; Parker and Appel [67]; Lameijer et al. [65]; Šperka and Halaška [36]
Strategic workforce value	9	Cycle-time reductions enable staff transition to value-adding roles; progressive transformation models show benefits accumulate across convergent, diffusing, and divergent stages	Negoită and Borangiu [81]; Mohamed et al. [80]; Bunduchi et al. [37]; Šimek and Šperka [66]
Financial and audit quality	7	RPA reduces internal control material weaknesses but paradoxically reduces monitoring intensity; ROI ranges 200–400% in mature deployments; regulatory environment quality predicts audit efficiency gains	Ashraf [40]; Jaradat et al. [11]; Teixeira et al. [88]; Ylä-Kujala et al. [35]
Sectoral-specific outcomes	14	Healthcare: safety compliance and time reductions; procurement: workload and benchmarking gains; utilities: operational resilience; public services: administrative efficiency with relational trade-offs	Wetsiri and Paireekreng [21]; van Hoek et al. [71]; Martins et al. [85]; Germundsson et al. [93]
Methodological contributions	5	Investment appraisal frameworks; performance measurement protocols; maturity-benchmarking models linking implementation performance to benefit trajectories	Ylä-Kujala et al. [35]; Quille et al. [83]; Rautenstrauch and Moser [87]

3.3.4. D4: Challenges and Barriers

The corpus makes a particularly concentrated contribution to understanding RPA adoption barriers, with the 2024–2025 cohort extending earlier typological and acceptance-model evidence toward more theoretically sophisticated accounts of frictions, sensemaking, and structural impediment.

Typological foundations: The authors of Kraus et al. [45] provide the most comprehensive and cross-contextually grounded challenge taxonomy to date, drawing on interviews with RPA consultants across industries to develop a typology encompassing the following challenges: (1) process-related challenges (inadequate process design, poor documentation, unstable inputs); (2) technical challenges (system integration failures, maintenance burdens); (3) resource-related challenges (skills gaps, budget constraints, insufficient staffing); (4) psychological challenges (employee resistance, fear of displacement,

loss of role identity); and (5) coordinative challenges (IT–business boundary conflicts, process ownership ambiguity, governance fragmentation). Critically, their inductive analysis establishes that psychological and coordinative challenges—not technical ones—are the most systematically underestimated failure modes, a finding corroborated by Schlegel et al. [38], who demonstrate that human capability gaps, inadequate project management, and governance deficits are the leading non-technical failure drivers. The authors of Dunlap and Lacity [64] provide early practitioner evidence predating this taxonomy: BCBSNC encountered exactly these categories in their pioneering deployment, resolving tussles over IT governance authority, process ownership, and the management of displaced workforce—establishing that negotiated organizational resolution, not technical configuration, determines implementation stability.

Systemic frictions as a novel theoretical contribution: The authors of Alastalo and Lehto [44] introduce the concept of “frictions” as analytically distinct from discrete technical “break-downs”, based on an ethnographic study of a Finnish healthcare RPA bot. Frictions—arising from the intersection of legislative constraints, austerity mandates, organizational hierarchies, and cost-saving pressures—are persistent structural impediments that continuously degrade bot performance without causing outright failure, and that standard failure-analysis frameworks miss because they focus on discrete error events rather than chronic performance erosion. This contribution extends the Kraus typology by identifying a category of impediment that is not addressable by technical maintenance but requires institutional and regulatory governance responses.

Acceptance and adoption barriers—multi-framework evidence: The corpus provides the most multi-theoretically and multi-nationally calibrated RPA acceptance evidence available. Foundational TAM and TOE work shows that adoption predictors are context-dependent: The authors of Sharma et al. [94] find in Indian service-sector organizations that Centre of Excellence maturity and low complexity are significant adoption predictors, while scalability counterintuitively shows a negative influence—suggesting governance overhead concerns suppress adoption enthusiasm in organizations without mature CoE infrastructure. The authors of Katke and Babu [95] calibrate TAM to RPA’s repetitive-task context, finding that perceived ease of use and subjective norms but *not* perceived usefulness predict adoption intention in IT and BPO settings. The authors of Vollenberg et al. [96] show through grounded theory with public sector employees that institutional legitimacy—not individual utility—is the primary acceptance driver in bureaucratic settings, and that performance expectancy is framed around operational improvement rather than cost reduction, requiring a distinct governance framing for public sector deployments. The authors of Razak and Ismail [97] extend TAM with Innovation Diffusion Theory in Malaysian accounting contexts, finding that cultural and organizational enablers are necessary complements to perceived usefulness for adoption intention. The authors of E-Fatima et al. [74] use Simul8 simulation to identify operational accuracy, shelf-life management, and food safety concerns as the specific barriers moderating RPA adoption in beef supply chains—demonstrating that sector-specific barrier taxonomies are required beyond standard IS adoption frameworks. The 2024–2025 cohort introduces theoretically richer barrier accounts: Zubler et al. [43] quantify status quo bias and dehumanization concerns as the strongest adoption inhibitors, proposing loss-framing countermeasures; Tailor and Goyal [98] integrate Leavitt’s organizational model with TAM in the TARPA model for sustainable transport SMEs; and Mayr et al. [46] find that trust and transparency in the AI component are uniquely critical for IPA compared to standard RPA.

Critical and socio-critical perspectives: The authors of Techatassanasoontorn et al. [51] identify root metaphors—person, robot, and tool—through which employees make sense of software robots across multiple organizations, demonstrating that managers use sense-giving to steer interpretation toward adoption-supportive framings and establishing that metaphor governance is as consequential as technical training. The authors of Hristova [99] offer a critical political-economy perspective, arguing that RPA process diagrams function as instruments of labor abstraction extending scientific management to white-collar work—raising workforce governance questions about visibility, agency, and the distribution of automation’s gains that standard acceptance frameworks elide. The authors of Germundsson et al. [93] and Ranerup and Svensson [53] document that, in Swedish social services, legislative constraints and relational consequences of automation create persistent barriers that TAM and TOE instruments cannot capture.

The corpus advances barrier understanding from typological description to theoretical explanation: The authors of Kraus et al. [45] establish the foundational taxonomy identifying five barrier categories across process-related, technical, resource, psychological, and coordinative domains. Critically, psychological and coordinative barriers—not technical ones—are the most systematically underestimated failure modes, corroborated by Schlegel et al. [38] and early practitioner evidence from BCBSNC’s pioneering deployment [64].

Two theoretical contributions extend understanding beyond taxonomy: Alastalo and Lehto [44] introduce “frictions” as analytically distinct from discrete “breakdowns”—persistent structural impediments arising from legislative, austerity, and organizational constraints that chronically degrade performance without causing failure. This contribution identifies a barrier category requiring institutional and regulatory governance responses rather than technical maintenance. The authors of Zubler et al. [43] empirically quantify status quo bias and dehumanization concerns as the strongest adoption inhibitors, proposing loss-framing countermeasures grounded in prospect theory.

Acceptance predictors vary systematically by institutional context: In bureaucratic settings, institutional legitimacy rather than individual utility drives adoption [96]; in emerging economies, CoE maturity and low complexity predict success while perceived scalability paradoxically suppresses enthusiasm [94]; in social services, legislative constraints and relational consequences create barriers that standard TAM and TOE frameworks cannot capture [53,93]. The 2024–2025 cohort provides the most theoretically sophisticated barrier accounts: TARPA model integration with Leavitt’s organizational model for transport SMEs [98], and trust and transparency governance uniquely critical for IPA compared to standard RPA [46].

Governance absence emerges as quantifiable risk: The authors of Dogan et al. [100] identify governance absence as the second-most critical risk factor through fuzzy FMEA, confirming that structural governance gaps constitute a quantifiable risk vector. The authors of Cascais Brás et al. [10] document governance failure as the primary explanation for RPA initiatives stalling after initial pilots, while Ashraf [40] reveal an automation complacency paradox: reduced internal control weaknesses coupled with reduced monitoring intensity. Dahiyat [101] provide early conceptual grounding for the relationship between RPA and audit quality, arguing that automation of audit-supporting processes improves consistency and reduces human error, though it requires governance mechanisms to prevent over-reliance.

Table 4 synthesizes adoption barriers across 49 studies, mapping barriers to theoretical lenses and sectoral contexts.

Table 4. Synthesis of adoption barriers across 49 studies.

Barrier Category	<i>n</i>	Key Barriers	Theoretical Lens/Key Studies
Process-related	12	Inadequate process design; poor documentation; unstable inputs; process complexity; unstructured automation	Typological [45]; sociotechnical [20]; Process redesign [48]
Psychological and workforce	18	Status quo bias; dehumanization concerns; job insecurity; resistance to change; role identity loss; neuroticism and turnover intention	SQB theory [43]; sensemaking [51]; burnout [91]; insecurity [102]
Technical and integration	8	System integration failures; maintenance burden; security vulnerabilities; complexity; compatibility barriers	Risk [100]; TOE [103]; implementation [34]
Organizational and governance	17	Skills gaps; governance absence; IT-business boundary conflicts; process ownership ambiguity; CoE immaturity; expectation management failures	Skills [47]; governance [10]; assimilation [77]; hype [78]
Institutional and structural	14	Legislative constraints; systemic frictions; austerity mandates; cultural barriers; procedural rights concerns; institutional isomorphism	Frictions [44]; public sector [53,96]; tax [82]; labor critique: [99]
Sectoral-specific	12	Beef supply chain accuracy, shelf-life, safety; hospitality stakeholder alignment; global business services standardization; transport safety-risk perceptions	Supply chain [49,74]; hospitality [22]; GBS [76]; transport [98]

3.3.5. D5: Technology Integration

Technology integration constitutes a structurally unifying dimension across the corpus, reflecting the field-wide recognition that RPA's maximum value is realized not as a standalone tool but as a component of broader automation and analytics architectures.

Conceptual positioning: RPA between BPM and AI: The authors of Dey and Das [55] provide the foundational positioning, drawing on survey evidence from RPA producer and user communities to establish that RPA occupies a distinctive niche between traditional BPM—which re-engineers processes through IT systems—and AI-driven automation, which handles unstructured data and cognitive tasks. Their exploratory factor analysis of user expectations yields three components: reducing repetitive manual work, optimizing business processes, and achieving ROI associated with implementation. Producer and user communities differ significantly in their prioritization of FTE reduction versus process optimization, a divergence with implications for how integration architectures are designed and evaluated. The authors of Rutschi and Dibbern [62] provide a complementary conceptual contribution, distinguishing RPA robots (back-office, structured process transformation) from chatbots (front-office, conversational process automation) through their routine transformation framework—establishing that the two robot types require different integration architectures and different implementation approaches due to differences in routine knowledge capture and surrounding actor relationships.

Process mining as the analytical substrate: The authors of Šperka and Halaška [36] develop the PPAFR framework applying process mining to a loan application process, demonstrating that event-log-based candidate identification produces more objective and reproducible

selection than manual assessment. This integration trajectory—process mining feeding RPA candidate identification—represents one of the most methodologically mature integration patterns in the corpus, providing data-driven discovery that complements criteria-based scoring. The authors of Hartikainen et al. [50] operationalize the downstream side of this trajectory through a comprehensive business-driven monitoring data model—establishing the observability infrastructure required for enterprise-scale RPA portfolio governance.

IPA and AI-augmented automation: The integration of AI capabilities with RPA into Intelligent Process Automation represents the most strategically consequential trajectory in the corpus. The authors of Liévano-Martínez et al. [79] demonstrate that IPA combining RPA and AI delivers quality and throughput benefits in manufacturing that exceed rule-based RPA alone, establishing IPA as the integration tier beyond task automation for production environments. The authors of Mohamed et al. [80] show that combined automation of administrative HR processes improves both efficiency and decision-support quality. The authors of Mayr et al. [46] advance the understanding of IPA adoption through an extended UTAUT model, finding that trust and transparency in the AI component are uniquely important adoption determinants—and that human-AI interaction design becomes progressively more consequential as automation intelligence increases. The authors of Waduge et al. [49] frame IRPA adoption in supply chains through a suitability model encompassing accuracy, human involvement, and standardization and Causal Loop Diagram, positioning intelligent automation as a strategic tool requiring systemic rather than task-level analysis.

Blockchain and smart contracts: The authors of Zeng et al. [23] demonstrate an architecturally novel integration in modular construction supply chains: smart contracts provide event-triggered contractual logic that automatically initiates RPA workflows, while RPA executes the resulting data processing across disparate systems. Their MSB-based triggering architecture achieves traceability and coordination benefits that neither technology delivers independently, validating the dual-engine model as a viable enterprise architecture pattern for contract-intensive, multi-party supply chains.

Three integration trajectories characterize RPA's evolution from standalone automation to enterprise architecture component: First, process mining integration provides data-driven candidate identification and portfolio monitoring [36,50]. Second, AI-augmented IPA extends automation to semi-structured data, with trust and transparency governance becoming uniquely critical [46,49,79]. Third, blockchain smart contracts enable event-triggered contractual logic that automatically initiates RPA workflows, validated in modular construction supply chains [23].

BPM redesign integration operationalizes RPA and BPM as unified improvement methodology: The authors of Santos et al. [48] demonstrate that applying BPM Redesign Heuristics before RPA deployment produces cycle time and workload reductions substantially greater than unstructured automation. The authors of McNamara et al. [63] provide a construction-calibrated integration maturity model progressing from rule-based RPA through AI-intensive document interpretation. The authors of Dumitru et al. [32] document how RPA integration into enterprise financial systems carries governance and reporting implications beyond operational efficiency, transforming management accountants' roles from data processors to interpreters.

Integration architecture choices govern productivity outcomes: Technology selection, configuration, and multi-bot orchestration decisions directly influence value realization [42,69,72]. Systematic performance measurement infrastructure forms the analytical substrate for integration value assessment [83]. Public-sector integration demonstrates distinct patterns in smart city billing and utilities contexts [73,85].

3.3.6. D6: Governance and Organizational Impact

Governance and organizational impact constitutes the broadest thematic dimension in the corpus. The evidence spans governance model design, workforce transformation, public sector and institutional governance, and the financial and compliance consequences of automation.

Governance model archetypes and structural choices: The most consequential governance design decision—the choice between centralized, federated, and decentralized models—is empirically grounded by two complementary studies. The authors of Cascais Brás et al. [10] demonstrate through a mixed-method study across multiple organizations that governance gaps at the intersection of business, IT, and people dimensions are the primary explanation for RPA programs stalling after initial pilots. They document three archetypes: centralized (single CoE authority), decentralized (business-unit managed), and federated (central standards with distributed execution), finding that the federated model best balances consistency with agility for mid-to-large organizations. The authors of Kedziora and Penttinen [52] provide a teaching case from Nordea Bank that operationalizes this choice: Nordea’s hub-and-spokes CoE model, initially implemented as a centralized structure, manages substantial automation at scale. The case also documents the governance tensions inherent in scaling—difficulties with component reusability, VDI management across multiple business units, and the challenge of coordinating maintenance across a geographically dispersed organization—establishing that governance architecture choice has direct consequences for implementation velocity, maintenance costs, and portfolio scalability. The case study also shows that license management—the ratio of robot licenses to business unit expectations—is a binding operational constraint that governance frameworks must address explicitly.

Governance as routine transformation: The authors of Rutschi and Dibbern [62] provide the most theoretically developed governance foundation: their routine transformation framework shows that governance must address not only technical bot design but the re-encoding of ostensive (abstract) and performative (enacted) routine elements. Governance frameworks that ignore the socio-organizational nature of routines—treating implementation as software configuration rather than as routine transformation—produce fragile automation that degrades with environmental change. The authors of Hartikainen et al. [50] operationalize this theoretically-grounded governance through a comprehensive monitoring data model, providing the observability architecture that enables systematic bot performance accountability.

Maturity-based governance: The authors of Rautenstrauch and Moser [87] propose a maturity model evaluating RPA governance performance across process, technology, people, and governance dimensions, enabling benchmarking and progression planning. The authors of Sobczak [77] identify the conditions governing organizational progression from initial adoption to sustained assimilation, finding senior management commitment, change management capability, and vendor governance as the primary determinants. The authors of Lacity et al. [58] develop action principles from the Maersk and Xchanging cases, bridging academic frameworks and operational decision-making through a practitioner governance vocabulary.

Workforce governance: from role redefinition to skills governance: Workforce governance is addressed through multiple complementary lenses. The authors of Šimek and Šperka [66] document how human–robot orchestration in an HR onboarding pilot requires careful role redefinition—establishing a single-dashboard process management interface that eliminates multi-system navigation—demonstrating that workforce governance must address both task substitution and task augmentation simultaneously. The BCBSNC case [64] establishes that workforce governance at scale requires advance planning: BCBSNC did

not anticipate the scale of workforce displacement, and the organization's commitment to its employees necessitated absorbing displaced staff through redeployment, establishing that human dignity considerations in workforce governance are not peripheral but organizationally defining. The authors of Schlegel and Kraus [47] provide structural evidence through content analysis of RPA-related job advertisements, documenting a persistent skills gap between available workforce competencies and RPA-enabled digital transformation requirements, with reskilling difficulty concentrated in the consulting-heavy RPA labor market. The authors of Techatassanasoontorn et al. [51] establish that workforce governance includes the symbolic and rhetorical management of automation: their identification of root metaphors (person, robot, tool) through which employees make sense of software robots shows that sense-giving is as consequential as technical training. The authors of Zhu and Kanjanamekanant [91] demonstrate that deployment breadth and depth differentially affect job autonomy and work intensification, and that governance of automation depth must actively protect employee autonomy to prevent burnout.

Public sector and institutional governance: Governance demands are profoundly shaped by institutional context. The authors of Vollenberg et al. [96] show that in public sector institutions, framing governance around service improvement rather than cost reduction is essential for employee acceptance, and that institutional legitimacy—the perception that RPA is organizationally sanctioned and democratically legitimate—is the primary acceptance driver. The authors of Johansson et al. [104] advance a normative framework arguing that public administration RPA governance must account for legal hierarchy, individual freedom, and democratic legitimacy beyond operational efficiency. The authors of Germundsson et al. [93] and Ranerup and Svensson [53] document governance failures in Swedish social services where automation of social assistance processes generates relational consequences—caseworker discretion constraints and client relationship degradation—requiring regulatory and professional governance responses beyond technical maintenance. The authors of Martins et al. [85] show that multi-municipal public utilities require coordinated cross-organizational governance, with process ownership boundaries as the critical resolved constraint.

Financial and compliance governance: The authors of Ashraf [40] document a governance paradox: automation reduces internal control material weaknesses at the firm level while simultaneously reducing audit committee monitoring intensity, creating an automation complacency risk that demands proactive governance countermeasures. The authors of Jaradat et al. [11] provide a governance antecedent model through PLS-SEM with Saudi Arabian internal auditors, finding that regulatory environment quality, vendor reputation, and data security infrastructure together predict audit efficiency gains, offering a governance design template aligned with national digitalization strategies. The authors of Teixeira et al. [88] show that institutional isomorphism—normative, mimetic, and coercive pressures—drives RPA adoption in Portuguese accounting firms, with implications for professional standards bodies seeking to govern technology adoption trajectories.

Risk governance: The authors of Dogan et al. [100] apply fuzzy FMEA to multiple robot deployment cases, quantifying governance absence as the second-most critical risk factor after sustainability challenges—providing a structured risk scoring approach for program management. The authors of Faúndez-Ugalde and Mellado-Silva [82] identify the rights dimensions of RPA in public tax administration as an emerging governance risk domain, arguing that automation must be governed to prevent disproportionate impacts on citizens' procedural rights.

Governance evidence organizes into five complementary dimensions beyond the structural and workforce topics already addressed: (1) maturity progression and assimilation—establishing conditions for organizational advancement from pilot to sustained deployment [58,77,87];

(2) participatory and change management approaches—addressing employee engagement, digital readiness, and psychological transition through participatory design and loss-framing strategies [43,102,105,106]; (3) sector-specific institutional responses—documenting governance adaptations in Malaysian accounting, Saudi auditing, hospitality stakeholder alignment, and transport safety-norm compliance [11,22,97,98]; (4) critical and sociotechnical perspectives—addressing labor abstraction concerns, epistemic and agentic object governance, and metaphor management [20,51,99]; and (5) risk, compliance, and value governance—covering investment appraisal, performance monitoring, supply chain risk modeling, and cross-organizational coordination [35,49,85,100].

The corpus demonstrates that governance maturity—not technical sophistication—determines sustained value realization: organizations progressing beyond pilot deployments require structural, process, and relational governance mechanisms that precede scale-out [10]. Federated governance models balancing central standards with distributed execution outperform purely centralized or decentralized approaches for mid-to-large organizations. Governance frameworks ignoring the socio-organizational nature of routines produce fragile automation that degrades with environmental change [62], while those addressing workforce autonomy protection, transparent communication, and progressive work transformation achieve broader and more durable implementation maturity [37,38].

4. Integrated RPA–BPM Framework

The SLR findings reveal that successful RPA–BPM integration is a multi-layered strategic undertaking spanning process, technology, people, and governance dimensions. The six thematic dimensions are not independent: process identification informs life cycle planning, technology integration shapes governance requirements, and challenges require coordinated responses across strategic, operational, and organizational levels. This interdimensionality motivates an integrated framework.

Drawing on the SLR synthesis, on established BPM frameworks [27], and on dynamic capabilities theory [29], we propose the Integrated RPA–BPM Framework comprising three pillars: **(I) Strategic Alignment**—ensuring RPA is grounded in organizational objectives and integrated with BPM strategy; **(II) Operational Execution**—translating strategic intentions into effective automation through rigorous life cycle management, technology integration, and change management; and **(III) Continuous Improvement**—sustaining and evolving RPA–BPM value through performance monitoring, maturity progression, and adaptive governance. The framework is presented in Table 5 and Figure 2.

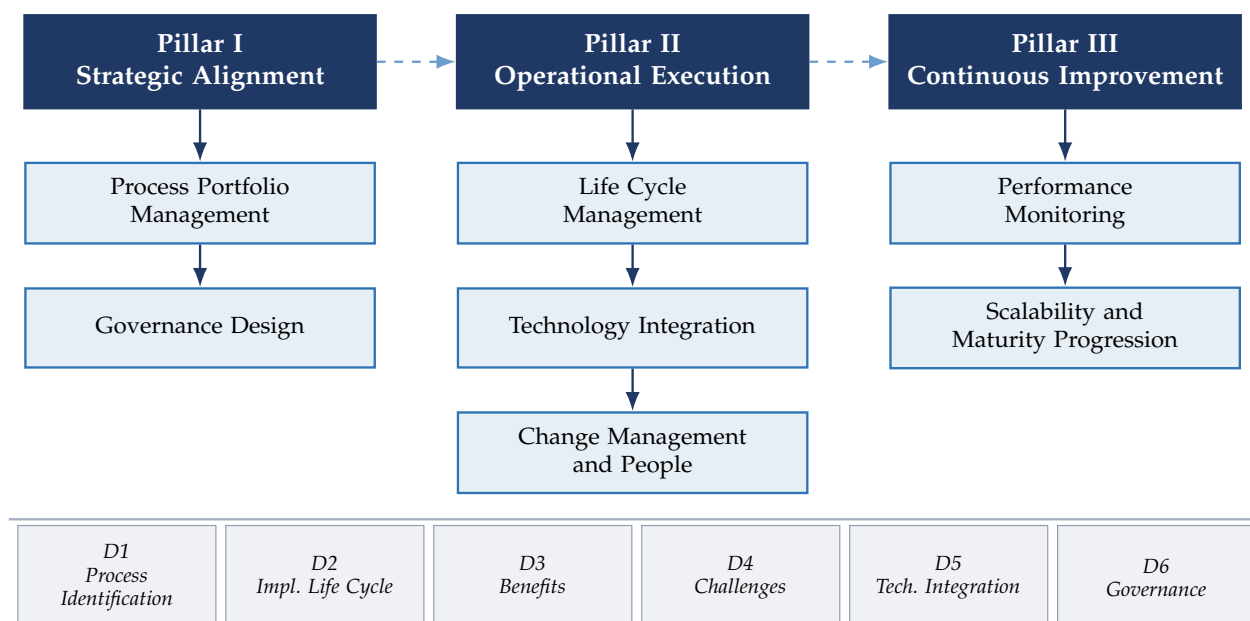
4.1. Pillar I: Strategic Alignment

Strategic alignment ensures that RPA deployment is driven by organizational strategy and aligned with the broader BPM agenda, rather than opportunistically triggered by isolated cost-reduction mandates. It encompasses two components.

Process Portfolio Management involves systematically building, maintaining, and prioritizing a pipeline of automation candidates. This goes beyond one-time process selection, requiring ongoing scanning for new opportunities, periodic reassessment of existing bots, and structured evaluation frameworks—MCDM or process mining-based—that align automation targets with strategic priorities [17,18,48]. The portfolio management perspective reflects the dynamic capabilities argument [29] that sustained advantage requires organizations to continuously sense, seize, and reconfigure their automation assets.

Table 5. Integrated RPA–BPM Framework: Three Pillars and Six Dimensions.

Dimension	Pillar I: Strategic Alignment	Pillar II: Operational Execution	Pillar III: Continuous Improvement
D1: Process Identification	Portfolio management; MCDM prioritization; automation-strategy alignment	Process analysis; requirements engineering; process mining-assisted discovery	Ongoing candidate pipeline; reassessment of deployed bots
D2: Implementation Life Cycle	Automation road map; digital transformation alignment; governance readiness	Six-phase life cycle; automated script synthesis; risk-managed testing	Iterative refinement; version control; life cycle maturity tracking
D3: Benefits and Outcomes	Business case; ROI/ROA projections; strategic benefit framing	KPI tracking: cycle time, error rate, throughput, FTE savings	Benefits realization reviews; value attribution; dashboards
D4: Challenges and Barriers	Risk assessment; stakeholder mapping; change readiness	Technical integration; employee engagement; IT readiness planning	Barrier monitoring; adaptation; lessons-learned documentation
D5: Technology Integration	Road map: RPA + Process Mining + AI/ML; vendor strategy	UI log capture; process mining-driven identification; IPA deployment	AI model integration; intelligent automation scaling
D6: Governance and Org. Impact	CoE design; governance model selection (centralized/federated/decentralized)	Role definition; access control; auditability; compliance management	Governance maturity; workforce reskilling; culture development

**Figure 2.** Integrated RPA–BPM framework: three pillars and six dimensions.

The governance design establishes structural, procedural, and relational mechanisms ensuring RPA delivers value at scale without creating operational risk. This includes defining the governance model (centralized, federated, or decentralized), establishing a Centre of Excellence (CoE) with clearly defined responsibilities, and designing compliance and auditability mechanisms satisfying regulatory requirements [10,50]. Crucially, governance design should precede—not follow—enterprise-wide scaling; organizations that delay governance establishment report higher failure rates and shadow-IT proliferation.

4.2. Pillar II: Operational Execution

Operational execution transforms strategic automation intentions into deployed bots delivering measurable process improvements. It encompasses three inter-related components.

Life Cycle Management: This applies structured methods across the full RPA life cycle [20,37]. Rigorous pre-automation process assessment is informed by structured scoring methods [17,18], disciplined effort estimation is performed using validated models [19], governance-ready deployment supported by monitoring infrastructure [50], and planned maintenance addresses not only technical degradation but the systemic frictions identified by Alastalo and Lehto [44].

Technology Integration: This positions RPA not as a standalone tool but as a component of an integrated automation and analytics architecture. The corpus demonstrates three maturing integration trajectories: BPM redesign combined with RPA for greater process improvement [48], blockchain smart contracts paired with RPA for supply chain traceability [23], and AI-augmented IPA for handling semi-structured and unstructured data [46,49]. Enterprise monitoring infrastructure [50] is the operational backbone enabling all three.

Change Management and People: This aspect addresses the human dimension of automation through transparent communication, early employee involvement in automation design, deliberate reskilling and redeployment programs, and the consistent framing of RPA as elevating human contribution [10,43]. The depth of change management investment is one of the strongest empirical predictors of RPA project success in the reviewed literature.

4.3. Pillar III: Continuous Improvement

The third pillar recognizes that RPA-enabled BPM is a dynamic, evolutionary endeavor. It encompasses two components.

Performance Monitoring and Analytics: This component sustains accountability through systematic KPI tracking across operational performance (cycle time, error rate, throughput), resource efficiency (bot utilization, FTE savings, processing cost), and organizational outcomes (employee satisfaction, return on automation). Process mining provides real-time, event-log-based visibility into bot execution, enabling proactive performance management [10,50].

Scalability and Maturity Progression: This component plans the organizational trajectory from initial pilots to enterprise-wide automation, and from rule-based RPA to AI-augmented IPA. Scaling introduces governance complexity, increased IT infrastructure demands, and heightened change management requirements that must be anticipated and managed. Dynamic capabilities theory frames this progression as an organizational learning journey building durable sensing, seizing, and reconfiguring capabilities that translate into sustained competitive advantage [29].

5. Discussion

5.1. Implications for Research

5.1.1. Positioning RPA Within Established BPM Theory

A central observation from this review is that the RPA–BPM literature has developed largely in parallel with—rather than integrated into—established BPM theory. The six core BPM elements of Rosemann and vom Brocke [27]—strategic alignment, governance, methods, IT, people, and culture—map directly onto the six thematic dimensions identified in this review, confirming a natural theoretical home for RPA within BPM scholarship. Future research should explicitly embed RPA within the BPM success factor literature [107], examining how RPA-specific factors (bot portfolio quality, CoE maturity) interact with classical BPM success conditions. Dynamic capabilities theory [29] provides a particularly promising lens: the ability to identify automation opportunities (sensing), deploy value-generating bots (seizing), and continuously reconfigure the automation portfolio (transforming) constitutes a distinctive dynamic capability amenable to empirical investigation. Resource-based theory [28] further suggests examination of VRIN conditions under which RPA constitutes a source of sustained competitive advantage.

5.1.2. Large-Scale Quantitative Research on Adoption and Outcomes

The reviewed corpus is predominantly qualitative and case-based, though the 2024–2025 wave of new publications introduces a stronger quantitative strand. Survey-based adoption studies now exist for general IPA [46], user resistance [43], technology acceptance [98], organizational adoption factors [103], and transport-sector workforce dimensions [98]. Prior quantitative work has examined RPA adoption using SEM and TOE frameworks in emerging economies [94], TAM-based adoption in public sector institutions [95,96], and competitive advantage pathways through RPA and digital business strategy [89]. Nevertheless, large-scale multi-industry studies remain absent. Future research should develop and validate measurement instruments for key RPA constructs—adoption intention, perceived benefits, implementation barriers, governance maturity, and performance outcomes—grounded in frameworks such as the Technology Acceptance Model [108] and the Information Systems Success Model [109]. The emerging TARPA model [98] and UTAUT extensions for IPA [46] provide starting points for instrument development. Structural equation modeling studies across diverse industries and institutional contexts would substantially advance the field’s generalizability. Particular attention should be given to non-adopter organizations—a conspicuous gap in the existing literature. Understanding the factors preventing adoption among potential adopters is essential for a complete theory of RPA diffusion and for informing policy and consulting interventions.

5.1.3. RPA Maturity Models, Governance Effectiveness, and Sociotechnical Theory

The literature documents a maturity trajectory from basic task automation to IPA, but this progression remains poorly theorized. The authors of Doolin et al. [20] offer the most developed process-theoretical account to date, mapping five implementation patterns (initiation, mobilization, configuration, adaptation, evaluation) and characterizing RPA as an object that evolves from a technical tool to an epistemic and agentic presence within organizations. The authors of Bunduchi et al. [37] provide a complementary stage model (convergent, diffusing, divergent) anchored in work transformation rather than IT implementation. Future research should synthesize these emerging sociotechnical accounts into maturity models specifying the capabilities, governance structures, and organizational conditions associated with each maturity stage, and identifying the dynamic capabilities required to progress between stages. Comparative studies of governance model effectiveness—examining how centralized, federated, and decentralized structures

perform across different organizational contexts—would extend IT governance theory into the RPA domain and provide practitioners with evidence-based governance design guidance. The risk governance literature [40,100] further highlights the need for dedicated governance maturity dimensions addressing operational risk, compliance, and the paradox of over-reliance on automation-as-control.

5.2. *Implications for Practice*

5.2.1. Begin with Processes, Not Technology

The most consistent practical lesson from the reviewed literature is that process quality and understanding must precede automation. Poor process selection is a primary cause of RPA failure [4,45], and automating a poorly designed process accelerates rather than remedies its deficiencies. Practitioners should invest in structured process assessment before deployment: the validated criterion sets of Yadav and Panda [17] and the RRPA scoring method of Bédard et al. [18] provide operationally ready evaluation tools applicable across industries. A comprehensive framework for selecting automation candidates based on 32 criteria validated through a Delphi study with RPA professionals provides additional pre-deployment rigor [33], while best practice guides for process description standardization support robust automation readiness [61]. The Lean DMAIC approach combined with RPA demonstrated by Huang et al. [39] illustrates how pre-automation improvement yields substantially greater outcomes than automation alone, while Santos et al. [48] show that BPM Redesign Heuristics applied before deployment deliver greater procurement gains than unstructured automation. Lean Six Sigma methods applied to existing RPA deployments further demonstrate that operational excellence techniques remain applicable in digitized environments [65].

5.2.2. Establish Governance Before Scaling

The transition from successful pilot to enterprise-wide RPA program is the most critical and most frequently problematic phase. The literature consistently demonstrates that scaling without governance infrastructure generates proliferating maintenance burdens, quality inconsistencies, shadow-IT problems, and compliance vulnerabilities [10,50]. Practitioners should design and instantiate governance structures—CoE formation, role definitions, selection criteria, change control, and bot monitoring frameworks—as a prerequisite to scale-out, not as a response to the problems that arise from it. Investment appraisal methods grounded in discounted cash flow modeling provide the economic governance foundation for scaling decisions [35], while systematic implementation frameworks supply the operational architecture for managing complexity as bot portfolios grow [34,38].

5.2.3. Address Adoption Barriers Through Theory-Grounded Strategies

The corpus identifies adoption barriers that go well beyond technical difficulty. The authors of Zubler et al. [43] show that dehumanization concerns and status quo bias are the strongest adoption inhibitors; and Alastalo and Lehto [44] show that systemic frictions arising from legislation, austerity, and organizational complexity persistently degrade bot performance. Practitioners should address these barriers proactively: through loss-framing and legitimation countermeasures for SQB [43], through transparent communication of role transformation rather than displacement for resistance [37], and through structural advocacy for regulatory and organizational conditions that reduce legislative frictions [44].

5.2.4. Treat the Workforce Transition as a Strategic Program

Employee resistance is the most commonly cited organizational barrier and is frequently underestimated in planning. The workforce implications of RPA—role transformation, task redefinition, reskilling requirements—require the same strategic attention

as the technical implementation [10,37]. The user resistance evidence of Zubler et al. [43] identifies subjective norms, dehumanization concerns, and status quo bias as the primary psychological obstacles; and Bunduchi et al. [37] document that organizations managing the transition through active goal alignment and manager facilitation achieve more sustained and broader work transformation outcomes. Prior research on human–bot co-working shows that deployment decisions—whether to extend automation depth or breadth—carry asymmetric implications for employee job autonomy and work intensification, and that governance of these dimensions shapes burnout risk and long-term adoption sustainability [91]. The skills gap between available workforce capabilities and RPA-driven digital transformation requirements underscores that reskilling is not a complementary benefit but a governance prerequisite [47]. Practitioners should establish workforce transition programs in parallel with technical deployment, involving employees early, communicating strategy transparently, and delivering concrete reskilling pathways demonstrating organizational commitment to workforce development. The hotel industry findings of Ge et al. [22] show that cultural context moderates which specific transition mechanisms are most effective, counseling against one-size-fits-all approaches.

6. Conclusions

This review reveals how RPA functions within BPM through systematic synthesis of 83 peer-reviewed articles (2015–2025, $\kappa = 0.82$). Six thematic dimensions characterize the RPA–BPM intersection: process identification, implementation life cycle, benefits, challenges, technology integration, and governance. These dimensions integrate into a conceptual three-pillar framework—strategic alignment, operational execution, continuous improvement—grounded in BPM theory and dynamic capabilities theory.

The corpus advances understanding beyond typological description toward theoretical explanation. Three conceptual contributions stand out: frictions as chronic-impediment concept [44]; empirical quantification of status quo bias as primary adoption inhibitor [43]; progressive work transformation models linking governance mechanisms to sustained maturity [37]. Technology integration patterns mature from standalone automation toward enterprise architectures combining process mining, AI-augmented IPA, and blockchain-triggered workflows. Pre-automation process quality emerges as the primary benefit moderator across all sectors.

Five research gaps warrant priority attention: multi-industry longitudinal adoption studies; validated sociotechnical maturity models specifying capability progression; comparative governance effectiveness evidence across centralized, federated, and decentralized models; non-adopter perspectives explaining resistance among potential adopters; cross-national institutional studies addressing legislative and cultural variance.

For practitioners, the framework provides structured navigation from process assessment and governance design through life cycle management and technology integration to performance monitoring and maturity progression. The evidence demonstrates that governance maturity—not technical sophistication—determines sustained value realization, and that organizations achieving durable implementation actively manage goal alignment, workforce autonomy protection, and transparent change communication.

7. Limitations of the Review

Several limitations should be acknowledged. First, the review is bounded by a 2015–2025 time frame and four databases, while these databases provide broad, high-quality coverage, they exclude all gray literature—practitioner reports, vendor publications, and government documents—that may contain practically relevant insights. Second, the restriction to peer-reviewed journal articles, while strengthening quality assurance, ex-

cludes conference proceedings, which are a significant venue in BPM and information systems research. Third, the English-language restriction may exclude relevant non-English scholarship. Fourth, content analysis in systematic reviews is subject to researcher interpretation; although a systematic coding approach was employed, thematic classifications reflect authorial judgment. Fifth, the integrated framework is derived from synthesis rather than empirical testing; its validity and utility in diverse organizational contexts remain to be confirmed through future investigation.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/technologies14040225/s1>, Table S1: PRISMA 2020 Checklist.

Author Contributions: Conceptualization, S.K. and P.S.; methodology, S.K.; investigation, S.K. and P.S.; data curation, S.K.; writing—original draft preparation, S.K.; writing—review and editing, S.K. and P.S.; visualization, S.K.; supervision, P.S. All authors have read and agreed to the published version of the manuscript.

Funding: Financial support for this study was provided by Mahasarakham Business School, Mahasarakham University, Thailand.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The PRISMA search log, screening records, and coding framework are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AI	artificial intelligence
BPM	Business Process Management
CoE	Centre of Excellence
ERP	enterprise resource planning
IPA	intelligent process automation
IRPA	intelligent robotic process automation
KPI	key performance indicator
MCDM	multi-criteria decision making
ML	machine learning
PRISMA	Preferred Reporting Items for Systematic Reviews
RBV	resource-based view
RPA	robotic process automation
SEM	structural equation modeling
SLR	systematic literature review
TAM	technology acceptance model
TOE	technology–organization–environment
UI	user interface
VRIN	valuable, rare, inimitable, non-substitutable

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